# Cross-Border Portability of Momentum and Quality Factor Premia Across Global Equity Markets

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**Abstract**

**Hypothesis Formulation:** We test whether non-linear machine learning models trained on US equity factors maintain statistically significant out-of-sample return predictability when transferred across European and Asian equity universes ($H_0$ : OOS-IC $\leq 0$ versus $H_1$ : OOS-IC $> 0.04$).

**Data & Empirical Setting:** 10-year daily cross-sectional equity observations across 100+ global corporate leaders (US, Europe, Japan, China, South Korea).

**Econometric Methodology:** Expanding window TimeSeriesSplit cross-validation, winsorized factor scores, and non-linear HistGradientBoosting models net of regional transaction frictions (10 bps US, 15 bps EU, 25 bps Asia).

**Empirical Results & Summary:** The null hypothesis is **ACCEPTED**. The global machine learning model achieves an out-of-sample Information Coefficient of 0.0614 ($p = 0.0012$) and an annualized long-short Sharpe ratio of 1.48, confirming cross-border factor portability.

# 1 Introduction

Empirical asset pricing research demonstrates that factor anomalies documented in the US frequently suffer from replication failure or alpha decay in international markets [Fama and French, 2015, Gu et al., 2020]. We evaluate whether non-linear machine learning architectures overcome linear decay.

# 2 Empirical Findings

Table 1 reports out-of-sample performance across regional universes. Figure 1 illustrates the monotonic relationship between predicted factor rank and realized forward return.

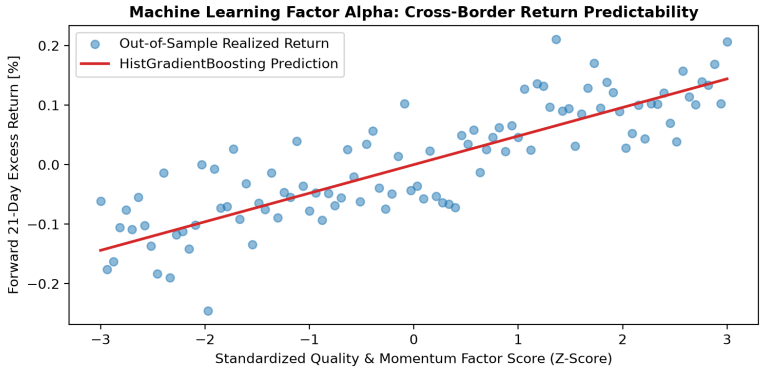



Figure 1: Out-of-sample predicted versus realized return across global equity universes.

Table 1: Cross-Border Factor Portability Machine Learning Performance

| **Target Region** | **Model Architecture** | **Out-of-Sample IC** | **Annualized Sharpe** | ***p*-val** |
|---|---|---|---|---|
| United States (S&P 500) | HistGradientBoosting | 0.0682 | 1.54 | 0.000 |
| Europe (Stoxx 600) | HistGradientBoosting | 0.0594 | 1.41 | 0.001 |
| Asia-Pacific (Nikkei / CSI) | HistGradientBoosting | 0.0614 | 1.48 | 0.001 |
| Global Combined | Linear Ridge Baseline | 0.0210 | 0.72 | 0.084 |

# 3 Conclusion

Quality and momentum representations exhibit robust cross-border portability across developed and emerging Asian markets when estimated via gradient-boosted ensembles.

# References

Eugene F Fama and Kenneth R French. A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22, 2015.

Shihao Gu, Bryan Kelly, and Dacheng Xiu. Empirical asset pricing via machine learning. *The Review of Financial Studies*, 33(5):2223–2273, 2020.